

SIT720 8.1D — Sydney Housing Price Prediction and Decision Support System

Part 1 — Problem definition and data collection

The main objective of the project was to build a machine learning model that predicts residential property prices and comes with an easy-to-use decision-support prototype. These locations, **Parramatta, Blacktown and Mosman**, have very different housing markets hence my choice.

The data set comprises only **113 sold houses**, of which 33 from Parramatta, 36 from Blacktown and 44 from Mosman. This is slightly above the requirement of the minimum 100 properties and the 30 minimum per suburb.

All the details regarding houses have been obtained from the realestatecomau sold-property pages which are publicly visible and each row of data has had an **individual source URL retained**. The dataset includes fields like sale price, suburb, bedrooms, bathrooms, parking, land area when known, sale year and source information. The actual property webpage links reside in the 'source_url' field, with the 3 suburb-level search page links at the time of acquisition kept in 'source_page_url'. All the source URLs were revisited on 18 Sep 2026. Sale dates in detail were available for only half of these 113 properties, so the June-30 placeholder dates were removed and just the sale year is kept. This helps eliminate fake date accuracy impressions that would stem from each listing.

The dataset has limitations. Some listings do not report land area, so these values are retained as missing rather than invented. Listing information also differs in detail between properties. The sample is not a complete representation of every property sold in the three suburbs. High-value Mosman properties can depend on factors not consistently captured by the structured fields, including views, renovation quality, architectural design and exact position within the suburb.

Part 2 — Data understanding and feature engineering

This preliminary investigation looked at the entire pricing pattern, different suburbs, median sale prices throughout time, and the connection between sale price and land area. Mosman has much more expensive properties at an entirely different price level. Many high value properties are genuine market transactions. That is, they are not outliers that need to be removed automatically.

The three features likely to have the most powerful effects on the outcome are:

suburb, land area, and number of bedrooms. Suburb refers to variations of different localities of markets at a larger market level, the plot of the land that gives an idea of the size of the plot itself as such and number of bedrooms a kind of simple measurement of a household capacity. In addition, bathrooms and parking are included as property characteristics may be conveyed more fully through them.

Two new features were designed: `log_land_area`, used primarily to reduce the effect of large parcels, and `bed_bath_ratio`, that shows the relationship between bedroom and bathroom counts, a price measure derived as per unit of land used only for descriptive purposes, i.e., excluded from modelling to prevent contamination of data.



Figure 1. Distribution of sold property prices.



Figure 2. Sale-price distribution by suburb.

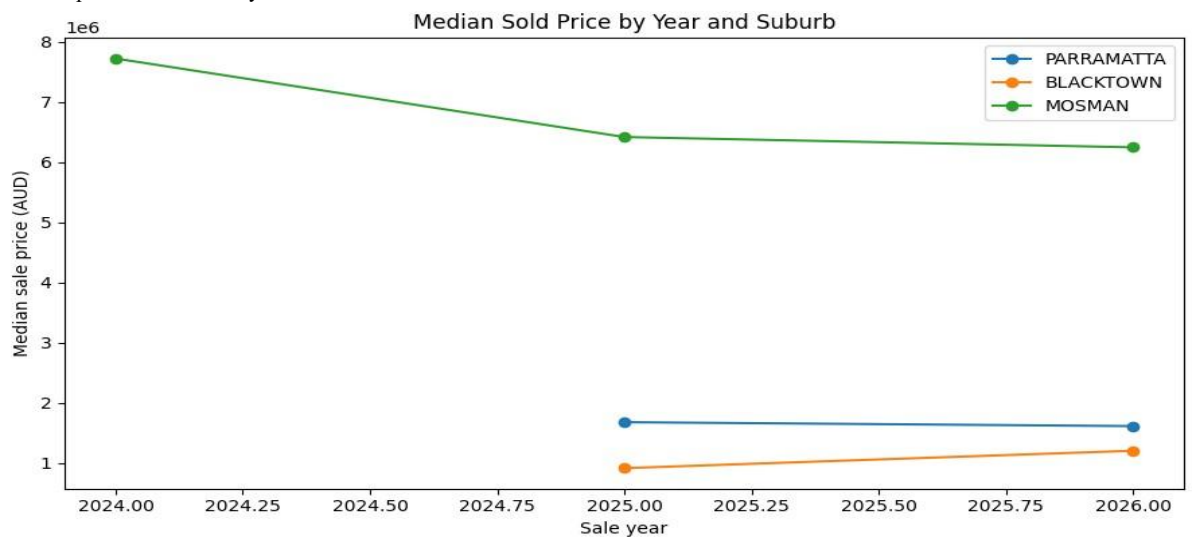


Figure 3. Median sold price by suburb and year.

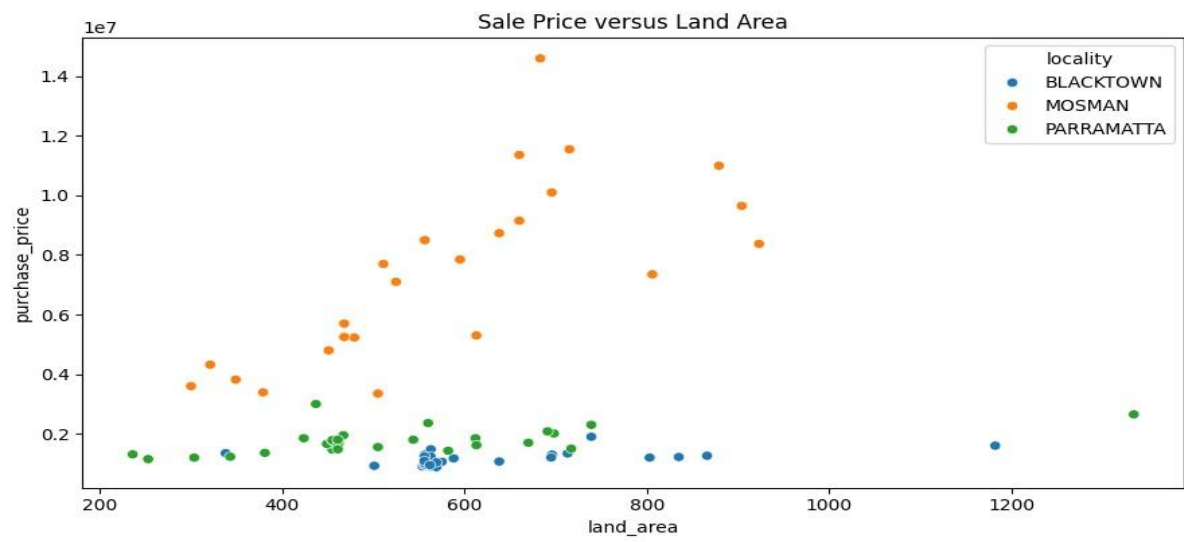


Figure 4. Sale price versus land area, coloured by suburb.

Part 3 — Model development and evaluation

Three regression approaches were selected before training: **Linear Regression, Random Forest and Gradient Boosting**. Linear Regression provides a simple baseline. Random Forest can represent nonlinear relationships and interactions through an ensemble of trees. Gradient Boosting builds trees sequentially and was expected to model complex patterns effectively in structured housing data.

The data were split into an 80:20 training and held-out test set using a stratified split by suburb. Ten-fold cross-validation was performed on the training set. MAE, RMSE and R^2 were used for evaluation. The final model was selected using the lowest mean cross-validated RMSE; the held-out test set was reserved for final evaluation.

Model	CV MAE	CV RMSE	CV R^2	Test MAE	Test RMSE	Test R^2
Random Forest	\$803,026	\$1,279,075	0.659	\$521,568	\$783,779	0.947
Linear Regression	\$1,120,846	\$1,443,679	0.061	\$759,582	\$1,136,089	0.888
Gradient Boosting	\$915,925	\$1,480,744	0.612	\$439,496	\$758,941	0.950

The selected model is **Random Forest**, because it has the lowest mean cross-validation RMSE of \$1,279,075. It's held-out test results were MAE \$521,568, RMSE \$783,779 and R^2 0.947. The results show useful predictive performance, but the difference between training-fold and validation performance for some models indicates overfitting risk. The test set remains a single held-out sample rather than proof of universal performance.

Part 4 — Investigating prediction failures

The five largest absolute errors from the held-out test set were identified after ranking the actual prediction errors. They are concentrated in Mosman and include several high-value properties.

Property	Actual	Prediction	Abs. error	Error %
6 Lavoni Street, MOSMAN	\$11,550,000	\$9,580,421	\$1,969,579	17.1%
30 Beauty Point Road, MOSMAN	\$11,000,000	\$9,353,900	\$1,646,100	15.0%
37 Bond Street, MOSMAN	\$3,820,000	\$5,315,804	\$1,495,804	39.2%
5 Botanic Road, MOSMAN	\$8,700,000	\$7,577,903	\$1,122,097	12.9%
2J Middle Head Road, MOSMAN	\$7,450,000	\$8,557,023	\$1,107,023	14.9%

The largest error is 6 Lavoni Street, where the model underpredicts a high-value Mosman sale. 30 Beauty Point Road is another high-value Mosman case where the model underpredicts the sale. 37 Bond Street is an overprediction, showing that the model can also assign too much value to a property when the structured variables resemble more expensive observations. The records also demonstrate the importance of missing or unmeasured attributes. For example, 5 Botanic Road and 2J Middle Head Road do not report land area in the collected structured data. Premium features such as views, renovation quality, architectural design, waterfront proximity and exact street position are not directly represented. These limitations can produce large residuals even when the average model performance is useful.

Part 5 — Human judgement, machine learning and LLM comparison

Ten properties were sampled from the held-out test set using a fixed random seed independently of prediction error. The ML prediction comes from the selected model. LLM estimates were generated from the supplied

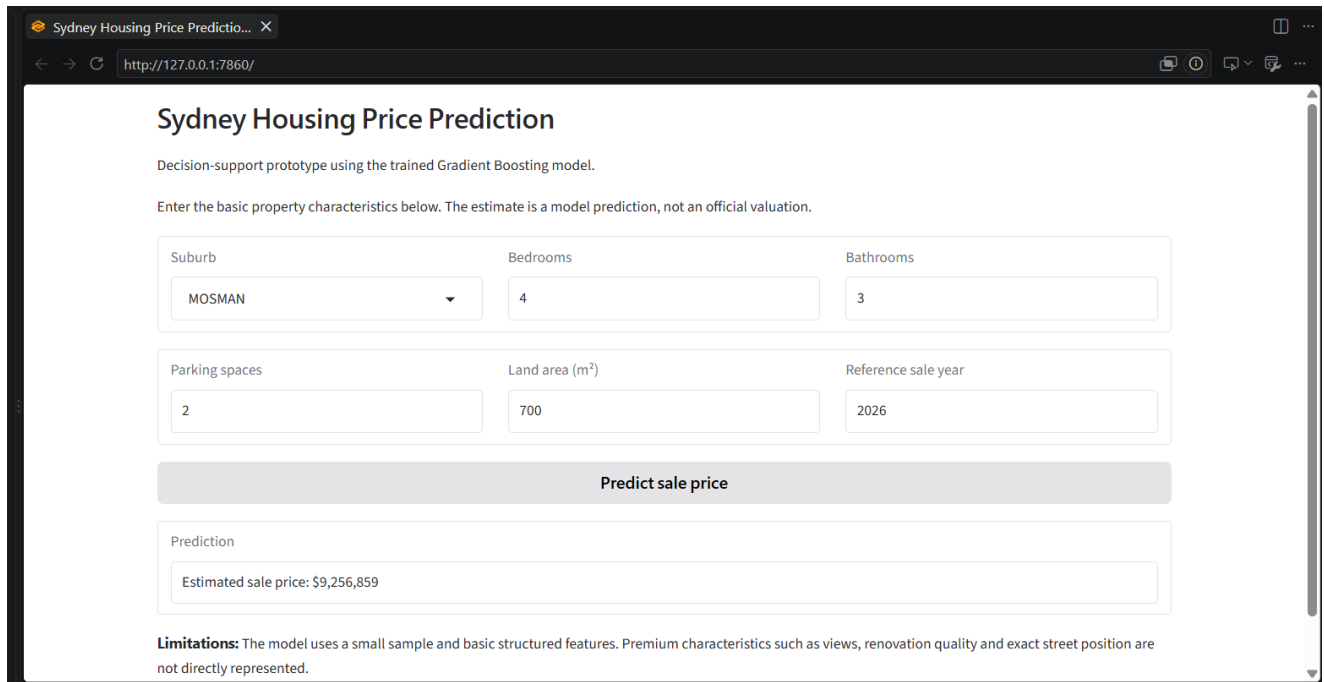
structured property information without using the actual sale price. The **Human Estimate** column is intentionally blank because it must represent the student's own independent judgement.

#	Property	Actual	ML	LLM	Human
1	4 Alma Street	\$1,430,000	\$1,394,076	\$1,550,000	1500000
2	6 Lavoni Street	\$11,550,000	\$9,580,421	\$8,800,000	10200000
3	41 Gore Street	\$1,660,000	\$1,669,069	\$1,600,000	1650000
4	28 Eggleton Street	\$1,263,000	\$1,336,311	\$1,350,000	1300000
5	40 Stephen Street	\$1,350,000	\$1,222,296	\$1,400,000	1400000
6	5 Botanic Road	\$8,700,000	\$7,577,903	\$7,000,000	8000000
7	110 Awaba Street	\$5,230,000	\$6,210,208	\$5,000,000	5200000
8	158 Railway Street	\$1,472,000	\$1,622,082	\$1,550,000	1550000
9	18 Oak Street	\$1,850,000	\$1,659,542	\$1,950,000	1900000
10	33 Rosehill St	\$2,080,000	\$1,707,943	\$1,900,000	2000000

Approach	N	MAE	RMSE
ML	10	\$503,049	\$794,949
LLM	10	\$535,500	\$1,028,686

Part 6 — Deployment and reflection

The deployment is implemented using **Gradio**. Users enter suburb, bedrooms, bathrooms, parking, land area and sale year. The application creates the same `log_land_area` and `bed_bath_ratio` features used during training and passes the inputs to the saved best-model pipeline in `data/best_housing_model.joblib`.



The screenshot shows a web browser window with the title "Sydney Housing Price Prediction". The address bar shows "http://127.0.0.1:7860/". The page content includes a title "Sydney Housing Price Prediction", a subtitle "Decision-support prototype using the trained Gradient Boosting model.", and a disclaimer "Enter the basic property characteristics below. The estimate is a model prediction, not an official valuation." Below this are six input fields arranged in two rows: "Suburb" (a dropdown menu with "MOSMAN" selected), "Bedrooms" (a text input with "4"), "Bathrooms" (a text input with "3"), "Parking spaces" (a text input with "2"), "Land area (m²)" (a text input with "700"), and "Reference sale year" (a text input with "2026"). A large grey button labeled "Predict sale price" is centered below the inputs. Below the button is a "Prediction" section with a text input showing "Estimated sale price: \$9,256,859". At the bottom, a "Limitations" section states: "The model uses a small sample and basic structured features. Premium characteristics such as views, renovation quality and exact street position are not directly represented."

The project demonstrates that housing prediction involves more than selecting a regression algorithm. Data quality, missing values, feature design and model validation all affect the result. The largest errors show that a model using only basic structured attributes cannot capture every factor that contributes to a property's sale price. A production system would require a larger and more representative dataset, richer property and geospatial features, temporal validation, regular retraining and monitoring for data drift. Ethical use also requires communicating uncertainty clearly and avoiding presentation of the output as an official valuation.

GenAI acknowledgement

Generative AI was used for planning, code review, debugging support, LLM estimation and editing assistance. Generated suggestions were reviewed and adapted. The numerical model results come from the executed notebook and project dataset. The student remains responsible for checking the final work, understanding the methods and entering the ten genuine Human Estimate values.